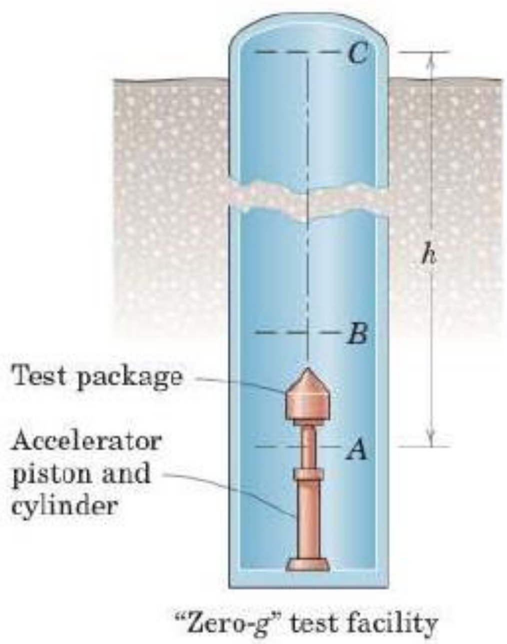
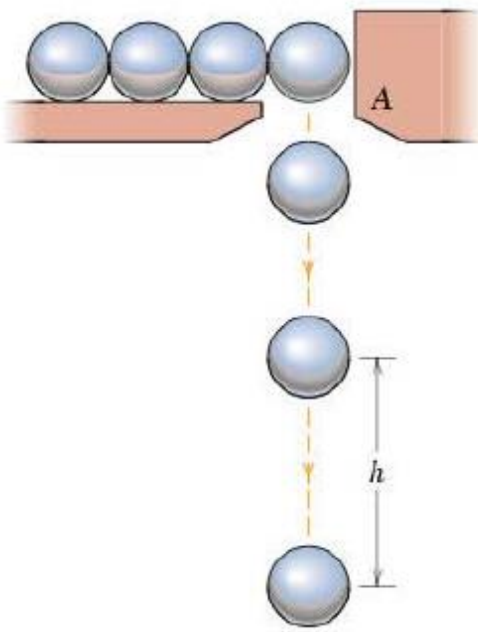
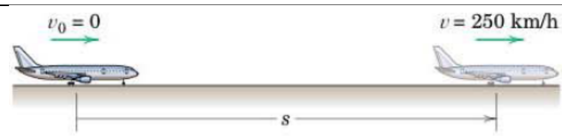
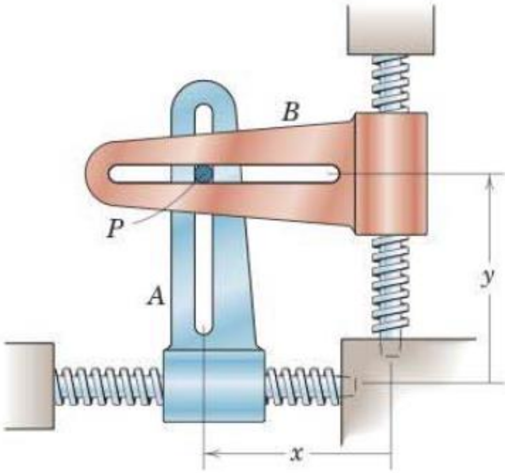
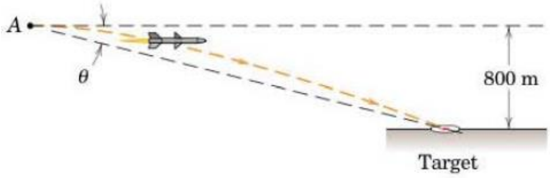
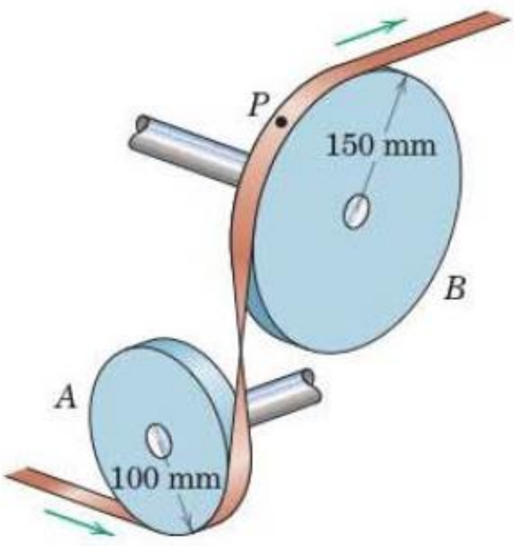
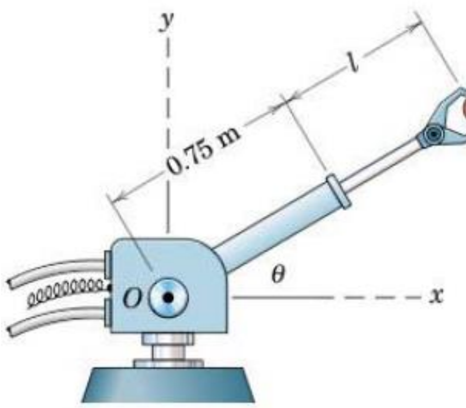
MEL1010 Engineering Mechanics
Tutorial 5 Kinematics of Particles

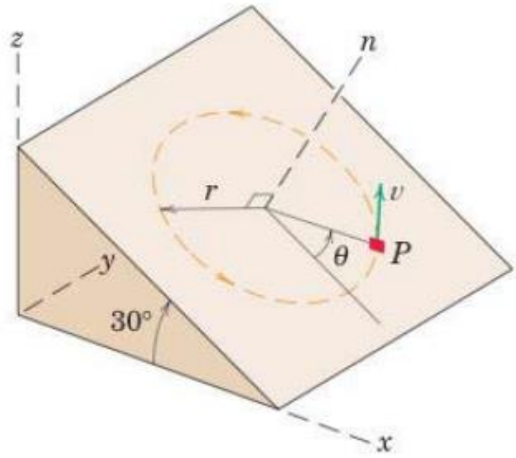
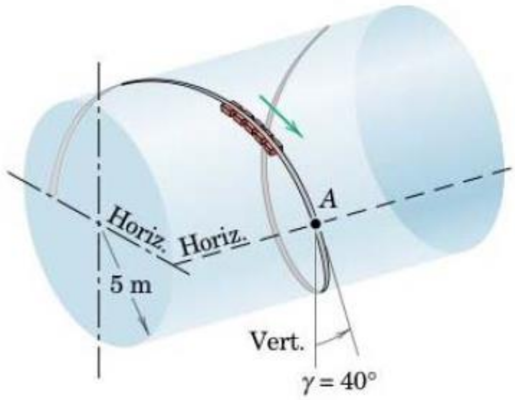
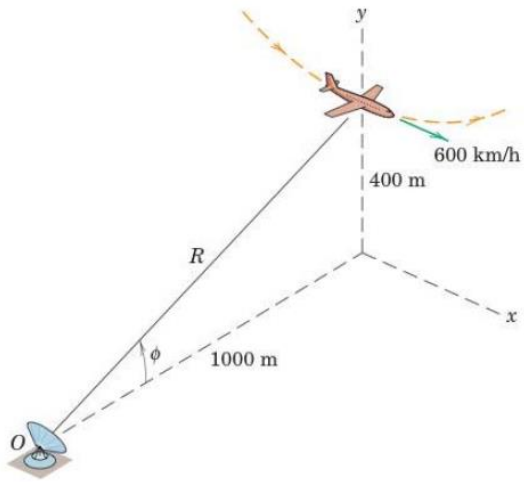
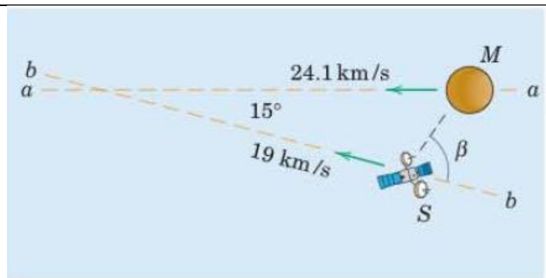
Tutorial Class: Thursday, 26th September 2025

Answer following questions. Mention valid assumptions wherever applicable

Questions about rectilinear motion, curvilinear motion using normal-tangential coordinates and polar coordinates, relative motion, space curvilinear motion, constrained motion

1	To test the effects of "weightlessness" for short periods of time, a test facility is designed which accelerates a test package vertically up from A to B by means of a gas-activated piston and allows it to ascend and descend from B to C to B under free-fall conditions. The test chamber consists of a deep well and is evacuated to eliminate any appreciable air resistance. If a constant acceleration of $40g$ from A to B is provided by the piston and if the total test time for the 'weightless" condition from B to C to B is $10\ s$, calculate the required working height h of the chamber. Upon returning to B, the test package is recovered in a basket filled with polystyrene pellets inserted in the line of fall. $Ans.\ h = 125.7\ m$	 "Zero-g" test facility
2	Small steel balls fall from rest through the opening at A at the steady rate of two per second. Find the vertical separation h of two consecutive balls when the lower one has dropped 3 meters. Neglect air resistance. $Ans.\ h = 2.61\ m$	
3	On its takeoff roll, the airplane starts from rest and accelerates according to $a = a_0 - kv^2$, where a_0 is the constant acceleration resulting from the engine thrust and $-kv^2$ is the acceleration due to aerodynamic drag. If $a_0 = 2\ m/s^2$, $k =$	

<table>
<tr><td></td><td>$0.00004\,m^{-1}$, and v is in meters per second, determine the design length of runway required for the airplane to reach the take-off speed of $250\ km/h$ if the drag term is (a) excluded and (b) included.
$Ans.\ (a)\ s = 1206\ m, (b)\ s = 1268\ m,$</td><td></td></tr>
<tr><td>4</td><td>The x- and y- motion of guides A and B with right-angle slots control the curvilinear motion of the connecting pin P, which slides in both slots. For a short interval, the motions are governed by $x = 20 + \frac{1}{4}t^2$ and $y = 15 - \frac{1}{6}t^3$, where x and y are in millimeters and t is in seconds. Calculate the magnitude of the velocity $\boldsymbol{v}$ and acceleration $\boldsymbol{a}$ of the pin for $t = 2\ s$. Sketch the direction of the path and indicate its curvature for this instant.
$Ans.\ \ v = 2.24\ mm/s,\ a = 2.06\ mm/s^2$
Note: Use Rectangular Coordinates</td><td>
</td></tr>
<tr><td>5</td><td>A rocket is released at point A from jet aircraft flying horizontally at 1000 km/h at an altitude of 800 m. If the rocket thrust remains horizontal and gives the rocket a horizontal acceleration of $0.5g$, determine the angle θ from the horizontal to the line of sight to the target.
$Ans.\ \ \theta = 30°$
Note: Use Rectangular Coordinates</td><td>
</td></tr>
<tr><td>6</td><td>The direction of motion of a flat tape in a numerical-control device is changed by the two pulleys A and B shown. If the speed of the tape increases uniformly from 2 m/s to 18 m/s while 8 meters of tape pass over the pulleys, calculate the magnitude of the acceleration of point P on the tape in contact with pulley B at the instant when the tape speed is 3 m/s.
$Ans.\ a = 63.2\ \mathrm{m/s^2}$
Note: Use normal-tangential coordinate</td><td>
</td></tr>
<tr><td>7</td><td>The robot arm is elevating and extending simultaneously. At a given instant, $\theta = 30°$, $\dot{\theta} = 10$ deg/s $=$ constant, $l = 0.5\ m$, $\dot{l} = 0.2\ m/s$, and $\ddot{l} = -0.3\ m/s^2$. Compute the magnitudes of the velocity $\boldsymbol{v}$ and acceleration $\boldsymbol{a}$ of the gripped part P. In addition, express $\boldsymbol{v}$ and $\boldsymbol{a}$ in terms of unit vectors $\boldsymbol{i}$ and $\boldsymbol{j}$.
$Ans.\ \ v = 0.296\ \mathrm{m/s},\ a = 0.345\ \mathrm{m/s^2}$
$\boldsymbol{v} = 0.064\ \boldsymbol{i} + 0.289\boldsymbol{j}\ \mathrm{m/s}$
$\boldsymbol{a} = -0.328\boldsymbol{i} - 0.1086\boldsymbol{j}\ \mathrm{m/s^2}$
Note: Use Polar Coordinate</td><td>
</td></tr>
</table>

<table>
<tr>
<td>8</td>
<td>The small block P travels with constant speed v in the circular path of radius r on the inclined surface. If $\theta = 0$ at time $t = 0$, determine the x-, y-, and z-components of velocity and acceleration as function of time.
Ans.
$$x = \frac{\sqrt{3}}{2} r \cos\frac{v}{r} t, y = r \sin\frac{v}{r} t, z = -\frac{1}{2} r \cos\frac{v}{r} t$$
$$\dot{x} = -\frac{\sqrt{3}}{2} v \sin\frac{v}{r} t, \dot{y} = v \cos\frac{v}{r} t, \dot{z} = \frac{1}{2} v \sin\frac{v}{r} t$$
$$\ddot{x} = -\frac{\sqrt{3}}{2}\frac{v^2}{r} \cos\frac{v}{r} t, \ddot{y} = -\frac{v^2}{r} \sin\frac{v}{r} t, \ddot{z} = \frac{1}{2}\frac{v^2}{r} \cos\frac{v}{r} t$$
Note: Space Curvilinear Motion, Rectangular coordinates</td>
<td>
</td>
</tr>
<tr>
<td>9</td>
<td>An amusement ride called the “corkscrew” takes the passengers through the upside-down curve of a horizontal cylindrical helix. The velocity of the cars as they pass position A is 15 m/s, and the component of their acceleration measured along the tangent to the path is $g \cos\gamma$ at this point. The effective radius of the cylindrical helix is 5 m, and the helix angle is $\gamma = 40°$. Compute the magnitude of the acceleration of the passengers as they pass position A.
Ans. $a = 27.5\ \mathrm{m/s^2}$
Note: Space Curvilinear Motion, cylindrical coordinates</td>
<td>
</td>
</tr>
<tr>
<td>10</td>
<td>At the bottom of a vertical loop in the x-y plane at an altitude of 400 m, the airplane has a speed of 600 km/h with no horizontal x-acceleration. The radius of curvature of the loop at the bottom is 1200 m. For the radar tracking at O, determine the recorded value of $\ddot{R}$ and $\ddot{\emptyset}$ for this instant.
$Ans. \ddot{R} = 34.4\ m/s^2, \ddot{\emptyset} = 0.01038\ rad/s^2$
Note: Space Curvilinear Motion, Spherical Coordinates</td>
<td>
</td>
</tr>
<tr>
<td>11</td>
<td>The spacecraft S approaches the planet Mars along a trajectory $b - b$ in the orbital plane of Mars with an absolute velocity of 19 km/s. Mars has a velocity of 24.1 km/s along its trajectory $a - a$. Determine the angle β between the line of sight $S - M$ and the trajectory $b - b$ when Mars appears from the spacecraft to be approaching it head on.
$Ans. \beta = 55.6°$
Note: Relative Motion</td>
<td>
</td>
</tr>
</table>

	Optional	
12	Under the action of force P, the constant acceleration of block B is $3\ \mathrm{m/s^2}$ to the right. At the instant when the velocity of B is $2\ \mathrm{m/s}$ to the right, determine the velocity of B relative to A, the acceleration of B relative to A, and the absolute veleocity of pont C of the cable. $Ans.\ v_{B/A} = 0.5\ \mathrm{m/s}, a_{B/A} = 0.75\ \mathrm{m/s^2}$ $v_C = 1\ \mathrm{m/s}, all\ to\ the\ right$ Note: Constrained Motion of Connected particles